AWS Basic to Advanced Bootcamp

Duration 4 Days

Day 1: AWS Essentials & Core Services

Concepts:

- Cloud Computing Overview (IaaS, PaaS, SaaS)
- AWS Global Infrastructure: Regions, AZs, Edge Locations
- AWS Free Tier and Pricing Model Overview
- AWS Console, CLI, and IAM Basics

Core Services:

- Amazon EC2: Launching, SSH Access, EBS Volumes
- EC2 Security Groups & Key Pairs
- Amazon S3: Buckets, Object Storage, Access Policies
- EBS vs S3 vs EFS

Hands-On Labs:

- Launch EC2 Instance with Apache/Nginx
- Upload and secure files on S3
- Use CLI to create and manage resources

Day 2: Networking, IAM & Database Services

Networking:

- Amazon VPC: CIDR, Subnets, Route Tables
- Internet Gateway, NAT Gateway, Bastion Host
- VPC Peering Basics

Identity and Access Management:

- IAM Users, Roles, Groups, and Policies
- Policy Simulator & IAM Best Practices
- STS and Role Assumption (Cross-Account Basics)

Database Services:

- Amazon RDS: Launch and Connect to MySQL/PostgreSQL
- DynamoDB: NoSQL Overview and Use Cases
- Aurora & Global Databases (Overview)

Hands-On Labs:

- Create custom VPC with public/private subnets
- Configure IAM roles and attach to EC2
- Launch RDS and access via EC2

Day 3: Serverless, Automation & Monitoring

Serverless Computing:

- AWS Lambda: Triggers, Permissions
- API Gateway Integration
- Step Functions (Intro)

Application Integration:

- Amazon SQS (Queue Service)
- Amazon SNS (Pub/Sub)
- EventBridge (Event-Driven Design)

Monitoring & Logging:

- Amazon CloudWatch (Logs, Alarms, Metrics)
- CloudTrail for Auditing
- AWS Config & Trusted Advisor

Automation:

- AWS CloudFormation Basics
- Introduction to AWS CDK (Preview)

Hands-On Labs:

- Deploy Lambda triggered by S3
- Setup REST API with API Gateway + Lambda
- Monitor EC2 via CloudWatch & send SNS alerts

Day 4: DevOps, Security & Advanced Architecture

DevOps Tools:

- CodeCommit, CodeBuild, CodeDeploy, CodePipeline
- Overview of ECS, EKS & Elastic Beanstalk
- CI/CD Strategy on AWS

Security & Cost Management:

- Shared Responsibility Model
- AWS KMS, Secrets Manager
- Cost Explorer, Budgets & Pricing Calculator

Well-Architected Framework:

- 5 Pillars: Security, Reliability, Performance, Cost, Operations
- Designing Multi-AZ & Highly Available Architecture

Certification Path & Wrap-up:

- AWS Certifications Overview (Practitioner → Architect Pro)
- Practice Questions & Exam Tips
- Career Path & Continued Learning Resources

Final Hands-On Project:

- Deploy a complete 3-tier architecture:
- Web Tier (EC2 + Load Balancer)
- App Tier (Lambda/API Gateway)
- DB Tier (RDS)
- Include S3, IAM, Monitoring & Security Policies